

G_2 –instantons on Joyce’s A_2 generalised Kummer construction

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Abstract

We apply Walpuski’s gluing theorem to Joyce’s Example 15, whose singular set has four transverse A_1 singularities and one transverse A_2 singularity. The A_2 ALE space carries a non-flat infinitesimally rigid tautological instanton. An explicit acyclic flat $\mathrm{PU}(3)$ connection and compatible equivariant ALE data give acyclic G_2 –instantons on a principal $\mathrm{PU}(3)$ –bundle $E \rightarrow Y$, where $(b^2, b^3) = (6, 17)$. Its adjoint bundle satisfies

$$p_1(\mathfrak{g}_E) = -\mathrm{PD}[S_{1,0}] - \mathrm{PD}[S_{1,1}] - 2\mathrm{PD}[S_A] \quad \text{in } H^4(Y; \mathbb{Q}).$$

1 Introduction

Let ϕ_0 be Joyce’s flat G_2 –structure on $\mathbb{T}^7$ and let $Y_0 = \mathbb{T}^7/\Gamma$ be the orbifold of [Joy96b, Example 15, pp. 350–351]. Its singular set consists of four copies of $\mathbb{T}^3$ with transverse model $\mathbb{C}^2/\mathbb{Z}_2$ and one copy of $\mathbb{T}^3/\mathbb{Z}_2$ with transverse model $\mathbb{C}^2/\mathbb{Z}_3$. Joyce resolves the former with Eguchi–Hanson spaces and the latter with an A_2 ALE space. There are two lifts of the residual involution to the A_2 space. We use the one acting trivially on H^2 .

We apply Walpuski’s gluing theorem [Wal13, Theorem 1.1, p. 2] with $G = \mathrm{PU}(3)$.

Theorem 1.1. *Let Y be the resolution of Joyce’s Example 15 obtained by using Eguchi–Hanson spaces at the four A_1 components and the first lift of the residual involution to an A_2 ALE space at the remaining component. There are $T_0 > 0$, a principal $\mathrm{PU}(3)$ –bundle $E \rightarrow Y$, and, for every $t \in (0, T_0)$, an acyclic G_2 –instanton A_t on E with respect to Joyce’s torsion-free G_2 –structure ϕ_t . If $S_{1,0}, S_{1,1}$ are the two A_1 components carrying non-flat ALE data and S_A is the A_2 component, then*

$$(1.1) \quad p_1(\mathfrak{g}_E) = -\mathrm{PD}[S_{1,0}] - \mathrm{PD}[S_{1,1}] - 2\mathrm{PD}[S_A] \quad \text{in } H^4(Y; \mathbb{Q}).$$

Since Y is simply connected, acyclicity excludes flatness. Moreover, Y has $(b^2, b^3) = (6, 17)$, and $\mathrm{Hol}(g_{\phi_t}) = G_2$.

We follow [Wal13, Sections 2–9].

2 G_2 –manifolds

Write $dx^{ijk} = dx^i \wedge dx^j \wedge dx^k$. In the coordinates of [Joy96b, (1)],

$$(2.1) \quad \phi_0 = dx^{127} + dx^{136} + dx^{145} + dx^{235} - dx^{246} + dx^{347} + dx^{567}.$$

If $(X, \omega_1, \omega_2, \omega_3)$ is hyperkähler and $(\delta^1, \delta^2, \delta^3)$ is an oriented orthonormal coframe on $\mathbb{R}^3$, then

$$(2.2) \quad \phi = \delta^{123} + \delta^1 \wedge \omega_1 + \delta^2 \wedge \omega_2 - \delta^3 \wedge \omega_3$$

is torsion-free, and every ASD connection on X pulls back to a G_2 -instanton; see [Wal13, Examples 2.8 and 3.4, pp. 5–6].

3 Gauge theory on G_2 -manifolds

Put $\Theta(\phi) = *_\phi \phi$. Let G be compact, connected and semisimple, with Lie algebra $\mathfrak{g}$, and let $E \rightarrow Y$ be a principal G -bundle. For $\psi = \Theta(\phi)$, a connection A is a G_2 -instanton if

$$(3.1) \quad F_A \wedge \psi = 0.$$

This is [Wal13, Definition 3.1, p. 6].

Put $\mathfrak{g}_E = E \times_G \mathfrak{g}$. In Coulomb gauge the linearised equation is controlled by the self-adjoint elliptic operator

$$(3.2) \quad L_A = \begin{pmatrix} 0 & d_A^* \\ d_A & *(\psi \wedge d_A) \end{pmatrix} : \Omega^0(Y, \mathfrak{g}_E) \oplus \Omega^1(Y, \mathfrak{g}_E) \longrightarrow \Omega^0(Y, \mathfrak{g}_E) \oplus \Omega^1(Y, \mathfrak{g}_E).$$

The connection is *acyclic* when L_A is invertible [Wal13, (3.10) and Definition 3.11, p. 8].

4 Joyce's generalised Kummer construction

4.1 The orbifold

On $\mathbb{T}^7 = \mathbb{R}^7 / \mathbb{Z}^7$ define

$$(4.1) \quad \alpha(x) = (x_3, x_4, x_5, x_6, x_1, x_2, x_7),$$

$$(4.2) \quad \beta(x) = (-x_1, \tfrac{1}{2} + x_2, -x_3, \tfrac{1}{2} + x_4, -x_5, \tfrac{1}{2} + x_6, -x_7),$$

$$(4.3) \quad \sigma_1(x) = (x_1, \tfrac{1}{2} + x_2, -x_3, \tfrac{1}{2} - x_4, -x_5, -x_6, x_7),$$

and $\sigma_2 = \alpha^2 \sigma_1 \alpha$, $\sigma_3 = \alpha \sigma_1 \alpha^2$. Joyce's relations imply

$$(4.4) \quad \Gamma = \langle \alpha, \beta, \sigma_1 \rangle \cong \mathbb{Z}_2 \times A_4$$

and Γ preserves (2.1); see [Joy96b, (41)–(43), Example 15]. The A_1 part of the singular set is obtained from the fixed tori of $\beta \sigma_i$. The action of $\langle \sigma_2, \sigma_3 \rangle$ on the sixteen tori fixed by $\beta \sigma_1$ is free and leaves four components in Y_0 .

The fixed set of α is

$$\{(u, v, u, v, u, v, w) : u, v, w \in \mathbb{R}/\mathbb{Z}\} \cong \mathbb{T}^3.$$

Its image is a single component $S_A \cong \mathbb{T}^3 / \mathbb{Z}_2$ with local model

$$(4.5) \quad (\mathbb{T}^3 \times (\mathbb{C}^2 / \mathbb{Z}_3)) / \mathbb{Z}_2, \quad (z_1, z_2) \sim (\zeta z_1, \zeta^2 z_2), \quad \zeta = e^{2\pi i / 3},$$

where the residual involution is

$$(4.6) \quad (y_1, y_2, y_3, [z_1, z_2]) \mapsto (\tfrac{1}{2} + y_1, -y_2, -y_3, [z_1, -z_2]).$$

This is [Joy96b, (44), pp. 350–351].

4.2 Resolution and perturbation

Let X_2 be the Eguchi–Hanson space and X_3 an A_2 ALE space asymptotic to $\mathbb{C}^2/\mathbb{Z}_3$. Joyce finds two lifts of (4.6) to X_3 . For the first,

$$(4.7) \quad \beta_X^* = \text{id} \quad \text{on } H^2(X_3; \mathbb{R}).$$

We use this lift. With four copies of X_2 and this copy of X_3 , Joyce’s Theorems 2.2.2 and 2.2.3 give a smooth simply connected 7-manifold Y with $(b^2, b^3) = (6, 17)$ [Joy96b, Example 15, p. 351]. Its torsion-free metrics have full holonomy G_2 by [Joy96b, Proposition 1.1.1, p. 330].

We use Walpuski’s notation for the resolution. On the scaled local piece $\tilde{T}_{j,t}$ set

$$(4.8) \quad \widehat{\phi}_{j,t} = \delta^{123} + t^2 \delta^1 \wedge \widehat{\omega}_{j,1} + t^2 \delta^2 \wedge \widehat{\omega}_{j,2} - t^2 \delta^3 \wedge \widehat{\omega}_{j,3}.$$

Cutting this off to ϕ_0 gives a closed positive form $\widetilde{\phi}_t$; see [Wal13, pp. 10–11]. Let r_t be the radial function of [Wal13, (4.13), p. 10].

Theorem 4.1 (Joyce–Walpuski). *There are $T, c > 0$ and $\eta_t \in \Omega^2(Y)$ such that, for $t \in (0, T)$,*

$$(4.9) \quad \phi_t = \widetilde{\phi}_t + d\eta_t$$

is torsion-free and

$$(4.10) \quad \|d\eta_t\|_{L^\infty} \leq ct^{1/2}.$$

This is [Wal13, Theorem 4.16, p. 11], citing [Joy96a, Theorems A and B] and [Joy96b, Theorem 2.2.1]. Put $w_t(x) = t + r_t(x)$ and $w_t(x, y) = \min\{w_t(x), w_t(y)\}$, and use the weighted Hölder norms $C_{\delta,t}^{k,\gamma}$ of [Wal13, pp. 11, 21].

Proposition 4.2 (Walpuski). *For each $\gamma \in (0, 1)$ the constants in Theorem 4.1 can be chosen so that*

$$(4.11) \quad \|d\eta_t\|_{C_{0,t}^{0,\gamma}} \leq ct^{1/2}, \quad \|\Theta(\phi_t) - \Theta(\widehat{\phi}_{j,t})\|_{C_{0,t}^{0,\gamma}(\tilde{T}_{j,t})} \leq ct^{1/2}.$$

This is [Wal13, Proposition 4.20, p. 11]; its proof is omitted.

5 ASD instantons on ALE spaces

Let X be ALE asymptotic to $\mathbb{C}^2/\mathbb{Z}_m$, $m \geq 2$. A framed ASD connection is *infinitesimally rigid* if the L^2 kernel of

$$(5.1) \quad \delta_A = d_A^* \oplus d_A^+ : \Omega^1(X, \mathfrak{g}_E) \longrightarrow \Omega^0(X, \mathfrak{g}_E) \oplus \Omega^+(X, \mathfrak{g}_E)$$

vanishes; see [Wal13, Definition 5.19, p. 19].

For a character χ of $\mathbb{Z}_m$, let $\mathcal{R}_\chi$ be the corresponding tautological line bundle and A_χ its canonical ASD connection. Kronheimer–Nakajima prove that the tautological connection associated with every unitary representation of $\mathbb{Z}_m$ is infinitesimally rigid [KN90, Lemma 7.1]; we use the formulation [Wal13, Theorem 5.20, p. 19].

5.1 The two local $\mathrm{PU}(3)$ instantons

On X_2 , let χ be the non-trivial character of $\mathbb{Z}_2$ and set

$$(5.2) \quad B_1 = \mathbb{P}(A_0 \oplus A_\chi \oplus A_\chi) \quad \text{on} \quad \mathbb{P}(\mathcal{R}_0 \oplus \mathcal{R}_\chi \oplus \mathcal{R}_\chi).$$

It is asymptotic to the projective class of

$$a = \mathrm{diag}(1, -1, -1) \in \mathrm{SU}(3).$$

On X_3 , write $\chi_j(1) = \zeta^j$ and set

$$(5.3) \quad B_2 = \mathbb{P}(A_0 \oplus A_{\chi_1} \oplus A_{\chi_2}).$$

Its boundary holonomy is $\mathrm{diag}(1, \zeta, \zeta^2)$, hence, after a Fourier change of framing, the projective class of

$$(5.4) \quad r = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathrm{SU}(3).$$

Proposition 5.1. *The connections B_1 and B_2 are infinitesimally rigid. With the inner product on $\mathfrak{pu}(3)$ induced by the negative Killing form,*

$$(5.5) \quad k(B_1) := \frac{1}{8\pi^2} \int_{X_2} |F_{B_1}|^2 = 1, \quad k(B_2) := \frac{1}{8\pi^2} \int_{X_3} |F_{B_2}|^2 = 2.$$

Moreover, B_2 admits a lift of the involution in (4.7) which preserves the connection and is the identity on the framing at infinity.

Proof. Rigidity follows from [Wal13, Theorem 5.20, p. 19]: the $\mathfrak{pu}(3)$ deformation complex is a parallel summand of the $\mathfrak{u}(3)$ complex.

For the charges, if V is a rank-three unitary bundle with Chern roots x_1, x_2, x_3 , then

$$(5.6) \quad p_1(\mathfrak{pu}(V)) = \sum_{i < j} (x_i - x_j)^2, \quad k = - \int_X p_1(\mathfrak{pu}(V))$$

for an ASD connection with the above normalisation. On X_2 , if $u = c_1(\mathcal{R}_\chi)$, then $\int u^2 = -1/2$ by [Wal13, Theorem 5.21, p. 19]; the roots $(0, u, u)$ give $k(B_1) = 1$. On X_3 , if $u_i = c_1(\mathcal{R}_{\chi_i})$, then

$$(5.7) \quad \left(\int_{X_3} u_i u_j \right)_{i,j=1}^2 = -C_{A_2}^{-1} = -\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The roots $(0, u_1, u_2)$ give $k(B_2) = 2$.

The group $H^2(X_3; \mathbb{Z})$ is torsion-free, and the boundary involution commutes with $\mathbb{Z}_3$. Thus (4.7) fixes each framed $\mathcal{R}_{\chi_j}$. Since β_X preserves orientation, its pullback of A_{χ_j} is ASD and is gauge-equivalent to A_{χ_j} : their curvature difference is exact, L^2 and ASD, hence has zero norm by Stokes, and $H^1(X_3) = 0$. Normalising the lift at infinity makes its square a parallel based gauge, hence the identity. Direct sum and projectivisation finish the proof. $\square$

6 Approximate G_2 -instantons

For a singular component S_j , let G_j be its transverse group, $\mu_j : \pi_1(T_j, x_j) \rightarrow G$ the local monodromy, and $K_j = \pi_1(T_j, x_j)/G_j$. Walpuski's Definition 6.3 requires a framed flat bundle (E_0, θ) , framed ASD data (E_j, A_j) asymptotic to $\mu_j|_{G_j}$, and a K_j -action on E_j covering the resolution action, preserving A_j , and asymptotic to μ_j [Wal13, pp. 20–21].

Compatible data determine a t -independent bundle $E \rightarrow Y$ and a connection $\tilde{A}_t$ obtained by cutting off θ to the scaled A_j . Write $\psi_t = \Theta(\phi_t)$. Walpuski's estimate is

$$(6.1) \quad \|F_{\tilde{A}_t} \wedge \psi_t\|_{C_{-2,t}^{0,\gamma}} \leq ct^{1/2};$$

see [Wal13, Proposition 6.7, pp. 21–23]. The same proposition gives

$$(6.2) \quad p_1(\mathfrak{g}_E) = - \sum_j k(A_j) \text{PD}[S_j] \quad \text{in } H^4(Y; \mathbb{Q}).$$

7 The model operator on $\mathbb{R}^3 \times X$

On $\mathbb{R}^3 \times X$, [Wal13, Proposition 7.4, p. 24] gives

$$(7.1) \quad L_A^* L_A = \Delta_{\mathbb{R}^3} + \begin{pmatrix} \delta_A \delta_A^* & 0 \\ 0 & \delta_A^* \delta_A \end{pmatrix};$$

Hence, for $-3 < \delta < 0$, [Wal13, Proposition 7.8, p. 26] identifies the weighted kernel of L_A with the L^2 kernel of δ_A ; it therefore vanishes for B_1, B_2 and, by [Wal13, Theorem 5.20, p. 19] applied to the trivial representation, for the trivial connection. Propositions 7.7 and 7.11 of [Wal13, pp. 25–26] give the rescaled comparison and uniform Schauder estimate.

8 Deforming to genuine G_2 -instantons

Let $L_t = L_{\tilde{A}_t}$. If θ is acyclic and every A_j is infinitesimally rigid, then [Wal13, Proposition 8.3, pp. 27–30] gives, for $-3 < \delta < 0$,

$$(8.1) \quad \|u\|_{C_{\delta,t}^{1,\gamma}} \leq c \|L_t u\|_{C_{\delta-1,t}^{0,\gamma}}$$

for all sufficiently small t . Together with (6.1), the contraction argument of [Wal13, Proposition 8.1, p. 27] yields an acyclic instanton

$$(8.2) \quad A_t = \tilde{A}_t + a_t, \quad \|a_t\|_{C_{-1,t}^{1,\gamma}} \leq ct^{1/2}.$$

9 The explicit $\text{PU}(3)$ example

9.1 The flat connection

Let $\Pi = \pi_1^{\text{orb}}(Y_0)$ be the affine deck group of the regular part. If τ_i translates x_i by one, then

$$(9.1) \quad 1 \longrightarrow \mathbb{Z}^7 \longrightarrow \Pi \longrightarrow \Gamma \longrightarrow 1, \quad \Pi = \langle \alpha, \beta, \sigma_1, \tau_1, \dots, \tau_7 \rangle \subset \text{Aff}(7).$$

In $SU(3)$ put

$$(9.2) \quad a = \text{diag}(1, -1, -1), \quad b = \text{diag}(-1, 1, -1), \quad c = \text{diag}(-1, -1, 1),$$

and let r be (5.4); the same letters denote their projective classes. Thus $rar^{-1} = b$, $rbr^{-1} = c$, $rcr^{-1} = a$. Define $\rho : \Pi \rightarrow PU(3)$ by

$$(9.3) \quad \rho(\tau_1) = a, \quad \rho(\tau_5) = b, \quad \rho(\tau_3) = c, \quad \rho(\alpha) = r,$$

and send $\tau_2, \tau_4, \tau_6, \tau_7, \beta, \sigma_1$ to 1.

Lemma 9.1. *The assignment (9.3) extends to a homomorphism.*

Proof. On the odd coordinate lattice, α cycles $e_1 \mapsto e_5 \mapsto e_3 \mapsto e_1$, exactly as conjugation by r cycles $a \mapsto b \mapsto c \mapsto a$. All sign changes disappear modulo two, and the half-translations in (4.2)–(4.3) occur only in even coordinates. Thus, if an affine element has coordinate-permutation part α^k and integral odd translation part (v_1, v_3, v_5) , its image is

$$a^{v_1} c^{v_3} b^{v_5} r^k,$$

with exponents reduced modulo two. This formula is multiplicative. $\square$

Let θ be the flat connection induced by ρ .

Proposition 9.2. *The connection θ is acyclic.*

Proof. By [Wal13, Proposition 9.2, p. 32], it is enough to show that

$$(9.4) \quad \mathfrak{pu}(3) \oplus (\mathbb{R}^7 \otimes \mathfrak{pu}(3))$$

has no non-zero Π -fixed vector. Since a, b, c are images of pure translations, their common fixed subspace in $\mathfrak{pu}(3)$ is the diagonal traceless Cartan subalgebra $\mathfrak{t}$. Conjugation by r cyclically permutes its diagonal entries, so $\mathfrak{t}^r = 0$.

For the tensor summand, translation invariance reduces the fibre factor to $\mathfrak{t}$. The fibre images of $\beta, \sigma_1, \sigma_2$ are trivial, whereas their tangent +1 eigenspaces are

$$\langle e_2, e_4, e_6 \rangle, \quad \langle e_1, e_2, e_7 \rangle, \quad \langle e_3, e_4, e_7 \rangle,$$

whose intersection is zero. Hence the tensor summand has no fixed vector either. $\square$

9.2 The four A_1 components

Put $g_0 = \tau_2^{-1} \tau_4^{-1} \beta \sigma_1$. Its fixed plane is parallel to $\langle e_2, e_3, e_5 \rangle$, and the fixed lifts have isotropy generators $\tau_1^p \tau_4^q \tau_6^\ell \tau_7^s g_0$ with $p, q, \ell, s \in \{0, 1\}$. Conjugation by σ_2, σ_3 identifies the q, ℓ choices, respectively. Thus, with $h_{p,s} = \tau_1^p \tau_7^s g_0$,

$$(9.5) \quad \pi_1(T_{p,s}) = \langle h_{p,s} \rangle \times \langle \tau_2, \tau_3, \tau_5 \rangle, \quad G_{p,s} = \langle h_{p,s} \rangle \cong \mathbb{Z}_2, \quad K_{p,s} \cong \mathbb{Z}^3.$$

The monodromies are

$$(9.6) \quad \rho(h_{0,s}) = 1, \quad \rho(h_{1,s}) = a, \quad (\rho(\tau_2), \rho(\tau_3), \rho(\tau_5)) = (1, c, b).$$

On $S_{0,0}$ and $S_{0,1}$ use the trivial connection. On $S_{1,0}$ and $S_{1,1}$ use B_1 . The constant diagonal gauge transformations b and c preserve (5.2); hence (9.6) defines the required $K_{p,s}$ -actions in Definition 6.3 of [Wal13].

9.3 The A_2 component

Set

$$(9.7) \quad \lambda_o = \tau_1 \tau_3 \tau_5, \quad \lambda_e = \tau_2 \tau_4 \tau_6.$$

Then

$$(9.8) \quad \pi_1(T_A) = \langle \alpha, \beta, \lambda_o, \lambda_e, \tau_7 \rangle, \quad G_A = \langle \alpha \rangle \cong \mathbb{Z}_3.$$

Here $K_A = \pi_1(T_A)/G_A$ is the Bieberbach group of S_A . From (9.3),

$$(9.9) \quad \rho(\alpha) = r, \quad \rho(\beta) = \rho(\lambda_o) = \rho(\lambda_e) = \rho(\tau_7) = 1,$$

because $acb = 1$. Use B_2 . Proposition 5.1 supplies the connection-preserving lift of β_X , the only non-trivial action on the ALE factor, and (9.9) gives the prescribed action at infinity. Its square matches $\beta^2 = \lambda_e$ because $\rho(\lambda_e) = 1$. Thus the A_2 data are compatible.

All local connections are infinitesimally rigid, and Proposition 9.2 gives the remaining hypothesis of Walpuski's Theorem 1.1. Sections 6–8 therefore produce acyclic G_2 -instantons A_t . Finally, (6.2) and (5.5) give

$$p_1(\mathfrak{g}_E) = -\text{PD}[S_{1,0}] - \text{PD}[S_{1,1}] - 2\text{PD}[S_A] \quad \text{in } H^4(Y; \mathbb{Q}),$$

which proves Theorem 1.1.

References

- [Joy96a] D. D. Joyce, *Compact Riemannian 7-manifolds with holonomy G_2 . I*, J. Differential Geom. **43** (1996), 291–328, doi:10.4310/jdg/1214458109.
- [Joy96b] D. D. Joyce, *Compact Riemannian 7-manifolds with holonomy G_2 . II*, J. Differential Geom. **43** (1996), 329–375, doi:10.4310/jdg/1214458110.
- [KN90] P. B. Kronheimer and H. Nakajima, *Yang–Mills instantons on ALE gravitational instantons*, Math. Ann. **288** (1990), 263–307, doi:10.1007/BF01444534.
- [Wal13] T. Walpuski, *G_2 -instantons on generalised Kummer constructions*, Geom. Topol. **17** (2013), 2345–2388, doi:10.2140/gt.2013.17.2345; revised version, 23 August 2014.